From classical energy estimates to a certified ground state: a modular Stetcu–Baroni filter pipeline and its cost for the J_1–J_2 chain

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(Dated: October 22, 2026)

Classical methods such as the density-matrix renormalisation group (DMRG) supply accurate ground-state energies, a gap estimate and a compact trial state, but a shallow circuit for that trial state keeps some weight on excited states. The projection filter of Stetcu, Baroni and Carlson removes it with one ancilla, controlled time evolution and post-selection. We give a generic, modular pipeline that takes energy estimates and an ansatz for any Hamiltonian written as a sum of Pauli strings, designs a cosine filter with a certified suppression ratio, chooses first- or second-order Trotter step counts from commutator bounds, and returns an a-priori bound on the infidelity of the prepared state together with a transparent gate count. The bound is a theorem conditional on the inputs (an interval for the ground energy, a lower bound on the gap and on the trial overlap), and every one of these conditions is an explicit input. We apply it to the open spin-$\frac{1}{2}$ J_1–J_2 chain with DMRG inputs and bond-dimension-χ matrix-product trial states, $N = 8$–48. In all 9 of 9 simulated circuits ($N \leq 14$) the certified bound held; it overestimates the measured infidelity by a median factor of 9, almost entirely in its Trotter term. An optimised filter beats the halving ("dyadic") schedule by a median 9× in expected CNOT count and a second-order product formula beats first order by a median 9×. At $N = 48$ near the critical coupling the certified circuit for infidelity 10^{-2} costs 9 CNOTs per successful preparation. Overestimating the gap is the one input error that breaks the guarantee in practice; a conservative gap costs little.

I. INTRODUCTION

A common route to a ground state on a quantum computer starts classically: an approximate method produces a trial state that can be compiled into a short circuit, and a quantum subroutine removes its residual excited-state weight. For one-dimensional and quasi-one-dimensional systems the natural classical engine is DMRG [**?** **?**], which yields the ground-state energy E_0, excited-state energies and a matrix-product state (MPS) whose low-bond-dimension truncations compile exactly into staircase circuits [**?**]. The quantum step can be phase estimation [**?**], a polynomial filter implemented by quantum singular-value transformation [**?** **?** **?**], or a product of cosine filters implemented with a single ancilla [**?** **?**]. The last option, the projection algorithm of Stetcu, Baroni and Carlson (SBC) [**?**], is attractive for early fault-tolerant hardware: each step is one controlled time evolution sandwiched between ancilla rotations and a measurement, the filter is designed classically, and the success probability is known in advance.

To use such a filter one has to answer three practical questions: which times and phases, how many Trotter steps, and how good the result is. SBC optimise the times and phases numerically for a model spectrum with a given gap; they do not give a bound on the error of the prepared state, and the Trotter error is treated separately [**?**]. This paper packages the full chain as a pipeline with typed inputs and a certificate:

1. *Inputs*: a Pauli-sum Hamiltonian H, energy estimates (an interval for E_0, a lower bound on the first excited energy E_1, optionally an upper bound on the spectrum) and an ansatz (its estimated ground-space weight γ and the gate cost of its circuit). Any classical method can supply them.
2. *Filter design*: three schedules, from the naive repeated cosine to an optimised design, each with a rigorous interval bound on its suppression ratio.
3. *Trotterisation*: first- and second-order commutator-scaling bounds [**?**], evaluated exactly on the support of each local commutator.
4. *Certificate*: an a-priori bound on the trace distance between the prepared state and the ground space, valid whenever the inputs are valid.
5. *Resources*: qubits, CNOTs, arbitrary rotations, a T-count estimate and the expected cost per successful preparation.

We then apply it to the open J_1–J_2 Heisenberg chain using DMRG, validate the certificate by statevector simulation for $N \leq 14$, and report resources for $N \leq 48$. The code is the `sbpipe` package in the accompanying repository; every number below is produced by the scripts named in Appendix **??**.

The contribution is methodological and quantitative rather than algorithmic: the filter is SBC's, the Trotter bounds are those of Childs *et al.* [**?**], and the certificate follows the standard triangle-inequality argument. What is new is a generic interface whose guarantees are stated in terms of the inputs, a closed-form "dyadic" energy schedule with $O(\Delta^{-1} \log \eta^{-1})$ total time, and a calibrated account of what this costs and where the bound is loose for a concrete, physically relevant model.

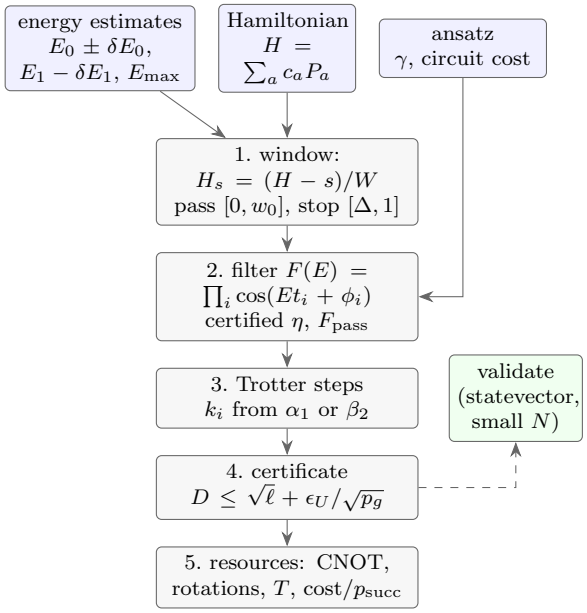


FIG. 1. The pipeline. Blue: inputs, which any classical method can supply. Grey: the stages of `refine()`. Green: the optional statevector check used in Sec. **??**.

II. THE PIPELINE

A. Inputs and the spectral window

The Hamiltonian is $H = \sum_a c_a P_a$ with real c_a and Pauli strings P_a, in a fixed order that is also the Trotter order, and with an optional grouping of terms (here: one group per bond). The pipeline assumes

$$E_0 \in [E_0^{\text{est}} - \delta E_0,\ E_0^{\text{est}} + \delta E_0], \quad E_1 \ge E_1^{\text{est}} - \delta E_1, \quad E_{\text{top}} \le E_{\max}, \tag{1}$$

where E_1 is the lowest level above the ground space that the trial state can overlap with. If no $E_{\max}$ is given, the Weyl bound $E_{\max} = \sum_g \lambda_{\max}(h_g)$ over the term groups h_g is used; it is rigorous and, for the antiferromagnetic J_1–J_2 chain, attained by the ferromagnetic state. With $s = E_0^{\text{est}} - \delta E_0$ and $W = E_{\max} - s$, the scaled Hamiltonian $H_s = (H - s)/W$ has its ground energy in the *pass band* $[0, w_0]$, $w_0 = 2\delta E_0/W$, and every other relevant level in the *stop band* $[\Delta, 1]$, $\Delta = (E_1^{\text{est}} - \delta E_1 - s)/W$.

The ansatz enters through γ, a lower estimate of its weight on the ground space, and through the gate cost of the circuit that prepares it. A statevector is needed only for validation.

B. The filter and three schedules

One SBC pulse [**?**] applies $\mathsf{H}_a\, e^{-itH_s \otimes Z_a}\, R_z^{(a)}(2\phi)\, \mathsf{H}_a$ to the system and an ancilla in $|0\rangle$; the ancilla-$|0\rangle$ block of this unitary is $\cos(H_s t + \phi)$. (SBC write the same block with $e^{-i(tH+\delta)\otimes Y_a}$.) Keeping outcome 0 after m pulses applies

$$F(H_s) = \prod_{i=1}^{m} \cos(H_s t_i + \phi_i), \qquad T = \sum_i |t_i|. \tag{2}$$

Two numbers characterise a filter: the pass-band floor $F_{\text{pass}} \le \min_{[0,w_0]} |F|$ and the suppression ratio $\eta = \sup_{[\Delta,1]} |F|/F_{\text{pass}}$. Both are certified, not sampled: since $|F'| \le L := \sum_i |t_i|$ and $|F''| \le L^2$, an interval branch-and-bound gives a rigorous upper bound on $\sup |F|$ over the stop band, and $F_{\text{pass}} = |F(0)| - Lw_0$ (the certifier is `floor.certify` of the repository). Given a target leakage ϵ_ℓ and γ, the design target is $\eta^2 \le \gamma\epsilon_\ell/[(1-\epsilon_\ell)(1-\gamma)]$ (Lemma **??**). We provide three schedules:

Uniform: m equal pulses $t = \pi/(1+\Delta)$, $\phi = 0$, so $F = \cos^m(Et)$ and $\sup_{[\Delta,1]} |\cos Et| = \cos\theta$, $\theta = \pi\Delta/(1+\Delta)$. Then $m \approx 2\ln(1/\eta)/\theta^2$ and $T \approx 2\ln(1/\eta)/(\pi\Delta^2)$: quadratic in the inverse gap. It is the baseline that any design should beat.

Dyadic: r repetitions of the halving ladder $t = \tau, 2\tau, \ldots, 2^{K-1}\tau$, $\tau = \pi/(1+\Delta)$, $\phi = 0$. This is the schedule SBC use for quantum-number projection [**?**], here applied to the energy. One block is a Dirichlet kernel,

$$\prod_{k=0}^{K-1} \cos(2^k E\tau) = \frac{\sin(2^K E\tau)}{2^K \sin(E\tau)}, \tag{3}$$

which equals 1 at $E = 0$ and is at most $\eta_b = 1/(2^K \sin\theta)$ on $[\Delta, 1]$ because $\sin(E\tau) \ge \sin\theta$ there. With $r = \lceil \ln\eta / \ln\eta_b \rceil$ blocks, $T = r\tau(2^K - 1)$; choosing $2^K \sin\theta \approx e$ gives

$$T_{\text{dyadic}} \approx \frac{e\,\tau}{\sin\theta} \ln\frac{1}{\eta} \ \approx\ \frac{e}{\Delta} \ln\frac{1}{\eta}, \tag{4}$$

the optimal $O(\Delta^{-1} \log \eta^{-1})$ scaling [**?**] with an explicit constant and no optimisation. The pipeline searches K and r and certifies the result.

Optimised: times and phases chosen to maximise $F(0)^2$ subject to the certified constraint on η at fixed (T, m), over a grid of $x = T\Delta/\pi$ and $m \in \{4, 6, 8\}$, picking the pair with the smallest $T^2/F(0)^2$. This is SBC's recipe for energy filters [**?**] with a certified constraint in place of a grid objective; it uses the SLSQP designer `floor.py` of the repository. Nonzero phases trade success probability ($F(0) < 1$) for a shorter T.

For each schedule the leakage share $\epsilon_\ell/\epsilon \in \{0.5, 0.25, 0.1, 0.05\}$ is scanned and the plan with the smallest expected CNOT count per success is kept.

C. Trotterisation

Each controlled evolution $e^{-it_i H_s \otimes Z_a}$ is approximated by k_i steps of the first-order (Lie) or second-order (Strang) product formula over the non-identity terms; the identity term of H_s becomes a single ancilla R_z per

pulse and is exact. For $B_a = \sum_{b>a} H_b$ the bounds of Childs *et al.* [**?** , Props. 15–16] are

$$\|S_1(t) - e^{-itH}\| \le \tfrac{t^2}{2}\alpha_1, \quad \alpha_1 = \textstyle\sum_a \|[B_a, H_a]\|, \tag{5}$$

$$\|S_2(t) - e^{-itH}\| \le t^3\beta_2, \quad \beta_2 = \tfrac{1}{12}\textstyle\sum_a \|[B_a,[B_a,H_a]]\| + \tfrac{1}{24}\textstyle\sum_a \|[H_a,[H_a,B_a]]\|, \tag{6}$$

so k steps over time t cost $\alpha_1 t^2/(2k)$ or $\beta_2 t^3/k^2$. Tensoring every term with Z_a leaves α_1 and β_2 unchanged because $Z_a^2 = 1$. For Pauli strings $[P_b, P_a] = 2P_bP_a$ when they anticommute and 0 otherwise, so each commutator is a short Pauli sum supported on a few qubits around term a; we evaluate its norm exactly on that support (dense for ≤ 8 qubits, Lanczos up to 12, triangle inequality beyond, which never occurred here). The constants are maximised over the forward and reversed term order, so they hold for either convention. A common step size, $k_i = \lceil t_i/\delta t \rceil$, minimises $\sum_i k_i$ for both orders at a given error budget (a Lagrange argument), and leaves the certified times untouched.

D. The certificate

Theorem 1. *Assume* (**??**) *and that the trial state has ground-space weight at least γ. Let the designed filter have certified η and F_{pass}, let ϵ_U bound the operator-norm error of the implemented pulses (Trotter error $\sum_i \epsilon_i$ plus $R\delta$ for R rotations each synthesised to accuracy δ), and let*

$$\ell = \frac{(1-\gamma)\eta^2}{\gamma + (1-\gamma)\eta^2}, \qquad p_g = \gamma F_{\mathrm{pass}}^2. \tag{7}$$

Then, conditioned on the ancilla outcomes 0, the prepared state $\tilde\rho$ satisfies

$$D(\tilde\rho, \Pi_0) \le \sqrt{\ell} + \frac{\epsilon_U}{\sqrt{p_g}}, \qquad 1 - F \le D^2, \tag{8}$$

where D is the trace distance to the normalised projection of the trial state on the ground space, $\langle H\rangle - E_0 \le (E_{\mathrm{top}} - E_0)D^2$, and the success probability is at least $(\sqrt{p_g} - \epsilon_U)^2$.

The proof (Appendix **??**) combines a leakage lemma for the exact filter with the observation that a vector error ϵ_U on an unnormalised vector of norm at least $\sqrt{p_g}$ moves its direction by at most $\epsilon_U/\sqrt{p_g}$. The pipeline sets the budget $\epsilon_U \le (\sqrt{\epsilon} - \sqrt{\ell})\sqrt{p_g}$, gives 90% of it to the product formula and the rest to rotation synthesis.

E. Resource model

All-to-all connectivity and the Clifford+R_z gate set are assumed. A term $cP\otimes Z_a$ with $|P| = w$ is a weight-$(w{+}1)$ Pauli rotation: $2w$ CNOTs in a ladder and one R_z. A Lie step costs one rotation per term; k Strang steps cost $k(2\Gamma - 2) + 1$ rotations for Γ terms because the two middle half-steps and the outer half-steps of adjacent steps merge. Each pulse adds one ancilla R_z. An arbitrary rotation at accuracy δ costs $3\log_2(1/\delta)$ T gates, the leading term of Ross–Selinger synthesis [**?**]. The cost of one attempt is the trial circuit plus the filter, and the expected cost per success is the attempt cost divided by the success probability, which we bound below by p_g (repeat-until-success; an early-abort protocol that measures after each pulse would be cheaper and is not modelled). The ladder CNOT count is an upper bound: qiskit's synthesis shares CNOTs between the XX, YY, ZZ terms of a bond and needs 8 instead of 12 per controlled bond per Lie step (checked in the test suite), so CNOT numbers below can be read as upper bounds that a compiler lowers by about one third.

III. APPLICATION: J_1–J_2 CHAIN WITH DMRG

A. Model and front end

We study the open spin-$\frac{1}{2}$ chain

$$H = J_1 \sum_{i=1}^{N-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_{i=1}^{N-2} \mathbf{S}_i \cdot \mathbf{S}_{i+2}, \tag{9}$$

$J_1 = 1$, at $J_2 = 0$ (Heisenberg), $J_2 = 0.2411$ (close to the critical point of the gapless-to-dimerised transition [**?** **?**]) and $J_2 = 0.4$ (dimerised), for $N = 8$ to 48. Its Pauli form has $3(2N-3)$ terms of weight 2. Terms are ordered by bond: nearest-neighbour bonds with even then odd left site, then next-nearest-neighbour bonds likewise.

The front end (`sbpipe/frontends/dmrg_quimb.py`) builds the MPO from the Pauli sum and runs two-site DMRG (quimb [**?**]) with bond dimensions up to 256 ($N \le 24$) or 128 ($N > 24$) and truncation cutoff 10^{-12}. E_1 is the lowest state of $H + \mu(S^z_{\mathrm{tot}} - 1)^2$, $\mu = 2$, i.e. the lowest triplet. The error bars $\delta E_0, \delta E_1$ are the energy change between the last two bond dimensions, floored at 10^{-10}; they are convergence heuristics, not bounds. The trial state is the variationally optimal MPS at fixed bond dimension $\chi \in \{1, 2, 4, 8\}$, and γ is its overlap with the reference MPS. An MPS of bond dimension 2^b is exactly a staircase of $N - b$ unitaries on $b + 1$ qubits; we cost each with the quantum Shannon decomposition bound $\frac{23}{48}4^n - \frac{3}{2}2^n + \frac{4}{3}$ CNOTs (3, 20, 100 for $n = 2, 3, 4$) [**?**] and $4^n - 1$ rotations.

For $N \le 14$ we compare with exact diagonalisation (ED). DMRG reproduces E_0 to 9, the first excited level of any symmetry to 9 (so the first excitation is the triplet in every ED case, as also found for $N \le 16$ in Ref. [**?**]), and the trial overlaps to 9. The Weyl bound equals the exact top of the spectrum to 9. Table **??** lists the inputs. The trial overlap decays roughly exponentially with N at fixed χ, the familiar orthogonality catastrophe, so the filter has more to do at larger N.

TABLE I. DMRG inputs: ground energy, gap $E_1 - E_0$ and trial overlap γ of the optimal bond-dimension-χ MPS. Source: `results/inputs.json`.

N	J_2	E_0	gap	$\gamma_{\chi=1}$	γ_2	γ_4	γ_8

FIG. 2. Certified bound versus measured value for all simulated plans. (a) Total infidelity; (b) Trotter distance between the Trotterised and the exact filtered state. Points below the dashed line respect the bound. Filled circles: second order; open squares: first order.

B. Validation of the certificate ($N \leq 14$)

For $N \in \{8, 10, 12, 14\}$, the three couplings, $\epsilon \in \{10^{-2}, 10^{-3}\}$, the three schedules and both Trotter orders we plan a circuit for the $\chi = 2$ trial state and, when its size permits, simulate it exactly: the ancilla-0 block of the Trotterised circuit is $\frac{1}{2}(e^{-i\phi}S(t) + e^{i\phi}S(-t))$ per pulse, applied to the trial statevector and compared with the ED ground state. Of 9 plans, 9 were simulated (the others are uniform or first-order plans with more than $\sim 10^6$ steps), and the certificate held in all 9 (Fig. **??**).

The bound is conservative by a median factor 9 (range 9–9) in infidelity. Splitting it shows where: the leakage term ℓ overestimates the exact filter's infidelity by 9–9× (median 9), because η is a worst case over the whole stop band while the trial state's excited weight sits mostly low in the band. The Trotter term overestimates the measured Trotter distance by a median 9× (range 9–9), consistent with earlier findings for this model [**?**]: commutator bounds are worst-case over all states, while the filtered state lives in the low-energy subspace, where Trotter errors are much smaller [**?**]. The measured success probability is 9–9 times the bound p_g.

Table **??** compares the schedules for one case. The uniform filter needs a total time an order of magnitude larger than the others, as its Δ^{-2} scaling predicts. The optimised filter is shortest; its nonzero phases lower p_g, which the cost criterion accounts for. Second order cuts the step count by one to two orders of magnitude at these sizes. Figure **??** shows the three filter functions.

FIG. 3. $|F(E)|$ of the three schedules for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$, second order. Dotted lines: certified ηF_{pass}. Grey: the gap region. The horizontal axis is linear below Δ and logarithmic above.

How much of the remaining gap could a better bound recover? For $N \leq 12$ we bisect a common scale factor on the certified k_i until the simulated infidelity just meets ϵ. The certified step count is 9–9× (median 9) larger than this oracle value (Table **??**). The oracle uses the exact ground state and is not a usable rule; it measures the price of certification at second order.

C. Resources up to $N = 48$

Without simulation the pipeline only needs DMRG, so we plan for N up to 48 (Fig. **??**, Table **??**). Over all sizes, couplings, targets and trial bond dimensions the second-order plan is 9–9× (median 9) cheaper than first order, and the optimised schedule is 9–9× (median 9) cheaper than the dyadic one at second order. For $N \geq 16$, $\chi = 4$, $\epsilon = 10^{-2}$ the expected CNOT count of the optimised second-order plan grows as N^9, N^9 and N^9 for $J_2 = 0, 0.2411, 0.4$ (first order: N^9, N^9, N^9). These exponents combine the shrinking gap ($T \propto 1/\Delta$), the growing commutator constants and number of terms, and the decaying overlap; they are fits over $N = 16$–48, not asymptotic statements.

D. What if the energy estimates are wrong?

The certificate is only as good as its inputs, so we perturb them for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$ (optimised, second order) and simulate the resulting circuit on the true trial state (Fig. **??**).

TABLE II. Plans for $N = 12$, $J_2 = 0.2411$, $\chi = 2$ trial. $T_{\text{phys}} = T/W$ in units of $1/J_1$; CNOT and T counts per attempt (filter) and expected per success (trial + filter, divided by p_g). "meas.": simulated infidelity of the planned circuit. Source: `results/validate.json`.

ϵ	schedule	order	m	T_{phys}	p_g	steps	CNOT (filter)	CNOT/success	T/success	meas. $1 - F$

FIG. 4. Expected CNOTs per successful preparation versus N (trial $\chi = 4$, $\epsilon = 10^{-2}$), including the trial circuit and the $1/p_g$ repetitions. Solid: second order; dashed: first order. Source: `results/scaling.json`.

TABLE III. Certified over oracle step count (second order, $\chi = 2$). Columns: $N = 8, 10, 12$ for each J_2. Source: `results/oracle.json`.

		$J_2 = 0$			$J_2 = 0.2411$			$J_2 = 0.4$		
ϵ	schedule	8	10	12	8	10	12	8	10	12

Gap. A gap assumed larger than the true one moves the stop-band edge above the first excitation, which then sits in the filter's transition region. In 9 of the overestimated cases the target was missed, starting at $f = 9$, with infidelity up to 9 at $f = 9$. Underestimating the gap is safe and cheap: assuming $f = 9$ costs at most $9\times$. Since DMRG gives an *upper* bound on E_1, an explicit margin δE_1 is the natural safeguard.

Ground energy. An error bar δE_0 widens the pass band and lowers F_{pass}; the cost stays flat until δE_0 is a sizeable fraction of the gap ($9\times$ at $\delta E_0 = 9$ gap). DMRG energies are far more accurate than this, so E_0 is never the limiting input here.

Overlap. Underestimating γ is safe (a stricter η) and

TABLE IV. Optimised plans for $\epsilon = 10^{-2}$, trial $\chi = 4$. Δ is the scaled gap, T_{phys} in $1/J_1$. Steps for first (1st) and second (2nd) order; CNOT and T counts for second order. Source: `results/scaling.json`.

N	J_2	γ	Δ	T_{phys}	p_g	steps (1st)	steps (2nd)	CNOT/attempt	CNOT/success	T/success

FIG. 5. Sensitivity to the inputs ($N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$). Circles: measured infidelity of the planned circuit on the true trial state. Squares: expected CNOT cost relative to the plan with exact inputs, multiplied by ϵ to share the axis. (a) Assumed gap $f\times$ the true gap; (b) error bar on E_0; (c) assumed overlap $f\times$ the true γ. Source: `results/sensitivity.json`.

costs 9× at $f = 0.25$. Overestimating it by 2% did not break the target here (9 failures), because the leakage term is itself loose.

E. Choosing the ansatz

The trial bond dimension trades circuit depth for overlap. Figure **??** and Table **??** show the expected total cost for $\chi = 1$ to 8. A product state ($\chi = 1$) has an exponentially small overlap and is the most expensive start; $\chi = 2$ or 4 is usually best, because beyond that the filter cost barely changes (it depends on γ only through ln and $1/p_g$) while the trial circuit grows as $4^{\log_2 \chi+1}$ per site. The best χ over the cases shown is in {9}. Stars mark trial states that already meet $1 - \gamma \leq \epsilon$ without a filter; when they exist they are much cheaper, which is the regime where direct MPS preparation should be preferred [**?**].

FIG. 6. Expected CNOTs per success versus trial bond dimension ($\epsilon = 10^{-2}$, optimised, second order). Stars: cost of the trial circuit alone when it already satisfies $1 - \gamma \leq \epsilon$. Source: `results/ansatz.json`.

TABLE V. Overlap γ and expected CNOTs per success for trial bond dimension $\chi = 1, 2, 4, 8$ ($\epsilon = 10^{-2}$, optimised, second order).

N	J_2	γ_1	CNOT	γ_2	CNOT	γ_4	CNOT	γ_8	CNOT

IV. DISCUSSION

What is guaranteed. Theorem **??** holds whenever its inputs are right. Of these, the spectral upper bound (Weyl), the filter certificate and the Trotter constants are rigorous here. The interval for E_0, the lower bound on E_1 and the overlap γ come from DMRG and are well-converged estimates, not bounds; for $N \leq 14$ ED confirms them, beyond that the certificate is conditional. The gap is the critical input: an overestimated gap was the only input error that broke the target in our tests. Rigorous lower bounds on E_1 from classical data (e.g. Temple-type bounds from the energy variance) would make the guarantee unconditional and plug into the same interface.

Where the cost goes. The certified cost is dominated by the Trotter term, which is two to three orders of magnitude loose. Second-order formulas recover most of the first-order looseness; bounds restricted to the low-energy subspace [**?**] or higher-order formulas are the obvious next lever. The leakage term is loose by about one order of magnitude, which would only be recovered with knowledge of the excited-state distribution of the trial state.

Is the filter worth it here? In one dimension, increas-

ing the bond dimension of a directly compiled MPS is a strong competitor at these sizes [**?**]; the filter's advantages are a certificate and a cost that depends on the trial state only logarithmically. The pipeline is generic in H, so the same comparison can be run for two-dimensional models where MPS compilation scales worse; we leave that to future work.

Limitations. Noiseless simulation; repeat-until-success rather than early abort; ladder CNOT counts (an upper bound, about 1.5× a compiler's); all-to-all connectivity; trial circuits costed by a generic decomposition bound rather than compiled; validation only up to $N = 14$.

Appendix A: Proof of Theorem ??

Write the trial state as $|\psi\rangle = \sqrt{\gamma'}|g\rangle + \sqrt{1-\gamma'}|\chi\rangle$ with $|g\rangle$ in the ground space (energy $e_0 \in [0, w_0]$ in scaled units), $|\chi\rangle$ orthogonal to it and $\gamma' \geq \gamma$.

Lemma 1 (leakage)**.** *For the exact filter, $v = F(H_s)|\psi\rangle$ has weight outside the ground space at most ℓ, and $\|v\|^2 \geq p_g$.*

Proof. $F(H_s)$ is a function of H_s, so $v = \sqrt{\gamma'}F(e_0)|g\rangle + \sqrt{1-\gamma'}F(H_s)|\chi\rangle$ with the two parts orthogonal. Every component of $|\chi\rangle$ has energy in $[\Delta, 1]$, so $\|F(H_s)\chi\| \leq \sup_{[\Delta,1]}|F| \leq \eta F_{\text{pass}} \leq \eta|F(e_0)|$. The excited weight is $x/(a+x)$ with $a = \gamma' F(e_0)^2$, $x = (1-\gamma')\|F\chi\|^2 \leq (1-\gamma')\eta^2 F(e_0)^2$; it is increasing in x and, after dividing by $F(e_0)^2$, decreasing in γ', which gives ℓ. Finally $\|v\|^2 \geq a \geq \gamma F_{\text{pass}}^2$. □

For pure states the trace distance is $\sin\angle$, so $D(\hat{v}, |g\rangle) = \sqrt{\ell'} \leq \sqrt{\ell}$ ($\hat{v} = v/\|v\|$).

Lemma 2 (implementation error)**.** *Let $B_i = \frac{1}{2}(e^{-i\phi_i}\tilde{S}_i(t_i) + e^{i\phi_i}\tilde{S}_i(-t_i))$ be the ancilla-0 block of the implemented pulse and $C_i = \cos(H_s t_i + \phi_i)$. Then $\|\prod_i B_i - \prod_i C_i\| \leq \epsilon_U$.*

Proof. $C_i = \frac{1}{2}(e^{-i\phi_i}e^{-it_iH_s} + e^{i\phi_i}e^{it_iH_s})$, so $\|B_i - C_i\| \leq \frac{1}{2}(\|\tilde{S}_i(t_i) - e^{-it_iH_s}\| + \|\tilde{S}_i(-t_i) - e^{it_iH_s}\|) \leq \epsilon_i$, where ϵ_i is the product-formula bound (symmetric in the sign of t) plus the synthesis error of the pulse's rotations. Since $\|B_i\|, \|C_i\| \leq 1$, telescoping gives $\|\prod B_i - \prod C_i\| \leq \sum_i \epsilon_i = \epsilon_U$. □

With $\tilde{v} = \prod B_i|\psi\rangle$, $\|\tilde{v} - v\| \leq \epsilon_U$. The distance from v to the line through $\tilde{v}$ is at most $\|v - \tilde{v}\|$, so $\sin\angle(v, \tilde{v}) \leq \epsilon_U/\|v\| \leq \epsilon_U/\sqrt{p_g}$. The triangle inequality for the trace distance gives (**??**). The success probability is $\|\tilde{v}\|^2 \geq (\|v\| - \epsilon_U)^2 \geq (\sqrt{p_g} - \epsilon_U)^2$. For the energy, the normalised state has weight at least $1 - D^2$ on the ground space and at most D^2 on levels $\leq E_{\text{top}}$, so $\langle H\rangle - E_0 \leq (E_{\text{top}} - E_0)D^2$. □

Appendix B: Reproducing the results

`python sbpipe_study/run_study.py all` runs the DMRG front end (cached), the validation (A), scaling (B), sensitivity (C), ansatz (D) and oracle (E) studies and writes `results/*.json`; `python sbpipe_study/make_figures.py` writes every figure, table and the numerical macros used in this text. `python -m pytest sbpipe/tests` runs the component checks (Pauli algebra, Weyl bound, Trotter bounds against dense matrix exponentials, the Dirichlet identity (**??**), filter certificates against a dense grid, gate counts against an explicit merged sequence and against qiskit, and an end-to-end certificate check).

[1] I. Stetcu, A. Baroni, and J. Carlson, Projection algorithm for state preparation on quantum computers, Phys. Rev. C **108**, L031306 (2023), arXiv:2211.10545.

[2] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, Theory of Trotter error with commutator scaling, Phys. Rev. X **11**, 011020 (2021).

[3] S. R. White, Density matrix formulation for quantum renormalization groups, Phys. Rev. Lett. **69**, 2863 (1992).

[4] U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, Ann. Phys. **326**, 96 (2011).

[5] S.-J. Ran, Encoding of matrix product states into quantum circuits of one- and two-qubit gates, Phys. Rev. A **101**, 032310 (2020).

[6] Y. Ge, J. Tura, and J. I. Cirac, Faster ground state preparation and high-precision ground energy estimation with fewer qubits, J. Math. Phys. **60**, 022202 (2019).

[7] L. Lin and Y. Tong, Near-optimal ground state preparation, Quantum **4**, 372 (2020).

[8] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, Quantum singular value transformation and beyond, in *Proc. STOC 2019*, p. 193.

[9] Y. Dong, L. Lin, and Y. Tong, Ground-state preparation and energy estimation on early fault-tolerant quantum computers via quantum eigenvalue transformation of unitary matrices, PRX Quantum **3**, 040305 (2022).

[10] K. Choi, D. Lee, J. Bonitati, Z. Qian, and J. Watkins, Rodeo algorithm for quantum computing, Phys. Rev. Lett. **127**, 040505 (2021).

[11] N. J. Ross and P. Selinger, Optimal ancilla-free Clifford+T approximation of z-rotations, Quantum Inf. Comput. **16**, 901 (2016).

[12] V. V. Shende, S. S. Bullock, and I. L. Markov, Synthesis of quantum-logic circuits, IEEE Trans. Comput.-Aided Des. **25**, 1000 (2006).

[13] K. Okamoto and K. Nomura, Fluid-dimer critical point in $S = 1/2$ antiferromagnetic Heisenberg chain with next nearest neighbor interactions, Phys. Lett. A **169**, 433 (1992).

[14] S. Eggert, Numerical evidence for multiplicative logarithmic corrections from marginal operators, Phys. Rev. B **54**, R9612 (1996).

[15] J. Gray, quimb: a python package for quantum information and many-body calculations, J. Open Source Softw. **3**, 819 (2018).

[16] B. Şahinoğlu and R. D. Somma, Hamiltonian simulation in the low-energy subspace, npj Quantum Inf. **7**, 119 (2021).

[17] Earlier analysis of the certified filter for this model in the same repository: `j1j2_filter/WRITEUP.md` and `simple_paper/paper.md`.